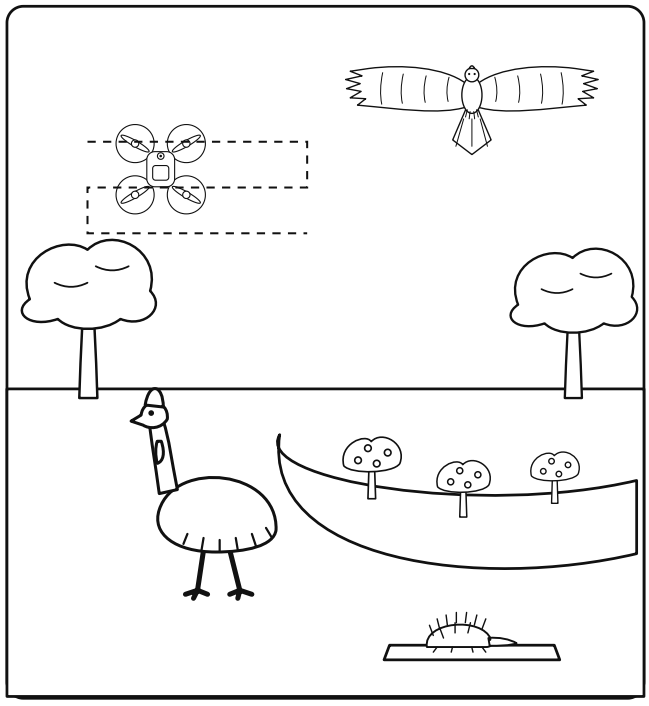
Printable pack
Level 4 · Year 6
No internet needed

AI FIRST NATIONS · DRONE KNOWLEDGE ON COUNTRY

NGARRAKAN NGARDIJI

LEVEL 4 · MAPPING COUNTRY · GOONDOI'S MAP



1 · Teacher guide and pathway
2 · Story to read aloud
3 · Rules
4 · Colour & Create
5 · Match-up
6 · Quiz
7 · Quiz answers
8 · Certificate

Created by John Fejo (Pop Pop), Founder & Chief Remote Pilot, AI First Nations · CASA ReOC CASA.ReOC.8979 · Darwin, Larrakia Country. Goes with Drone Class 301 (intro).

LEVEL 4 TEACHER GUIDE

Level 4: Mapping Country. The rangers need a map of pond apple weeds. Goondoi, the cassowary, shows Ngarrakan where the weeds grow, and Kiki plans her first mapping flight. Echidna stays safe under a log. This level teaches planning and flying a mapping mission.

Time	What happens	Page
5 min	Acknowledgement of Country. Invite an Elder or ranger to share the local names for the animals in today's story.	—
10 min	Read the story aloud, or play it on a screen. Stop at each scene and yarn about it.	2
10 min	Rules. Read them together and link each rule to the story.	3
15 min	Colour & Create, and the match-up.	4, 5
10 min	Quiz, marked together.	6, 7
10 min	Certificate. If flying is planned, run the pre-flight checklist with the instructor.	8

THE PATHWAY: SAME SIX STEPS, YEAR 3 TO YEAR 12

Every level: **1 Story · 2 Rules · 3 Colour & Create · 4 Match-up · 5 Quiz · 6 Certificate.**

Lvl	Years	Level and story	CASA focus
1	Year 3	**First Flight** The Pilot and the Spotter	Keep the drone in sight · no higher than 120 m · 30 m from people · daylight only · never over people.
2	Year 4	**Safe Flight** Storm Coming	Weather and wind limits · battery reserve · Return to Home · lost signal · keep clear of emergencies.
3	Year 5	**Eyes in the Sky** Crocodile Says Keep Back	Keep distance from wildlife · privacy · no filming people without asking · camera settings with a purpose.
4	Year 6	**Mapping Country** Goondoi's Map	Planned missions · overlap · battery planning · instructor holds the controls for automated flights.
5	Year 7 to 8	**Sharing the Sky** Fire on the Hill	Airports and 5.5 km · CASA-verified drone safety apps · NOTAMs · registration · give way to crewed aircraft.
6	Year 9 to 10	**AI on Country** Dugong and the Counting Machine	Data from commercial flights under a ReOC · storing data safely · human checks on AI results.
7	Year 11 to 12	**Careers in the Sky** Kiki Earns Her Wings	Operator accreditation (16+) · RePL · ReOC · registration · job safety analysis · client briefs.

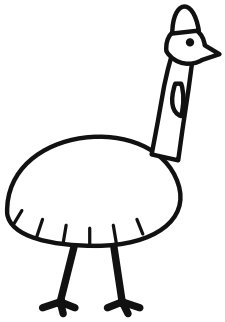
Flights in AI First Nations classes run under ReOC CASA.ReOC.8979 with the instructor in charge. Students under 16 always fly under supervision. Culture: Maps of Country are shared only with the people who own that Country.

PAGE 2 · STORY

GOONDOI'S MAP

Read this aloud to the class. On a screen, the same story plays as an animation.

1. The rangers need a map of pond apple, a weed that chokes the wetlands. Ngarrakan asks Goondoi, the cassowary, where it grows.
2. Kiki plans her first mapping mission on the tablet: a grid of straight lines over the swamp, with photos that overlap.
3. For automated flights, Pop Pop holds the controller. The drone flies up and back along the grid while KK spots.
4. Echidna is resting under a log. The drone stays high and steady, so Echidna isn't disturbed.
5. Back at base, all the photos are joined into one map, called an orthomosaic. The pond apple patches show up clearly.
6. The map goes only to the rangers and Traditional Owners, so they can treat the weeds. Goondoi's swamp can breathe again.



Yarn about it: How did Goondoi help Ngarrakan find the weeds? Why did the drone stay high over Echidna? What would you like to map on your Country, and who would you ask?

MAPPING COUNTRY

1 Plan the area with rangers and Traditional Owners
Agree on where to map, and which places stay off the map.

2 Fly a grid pattern
Mapping flights go up and back in straight lines, like mowing a lawn.

3 Overlap your photos
Each photo shares about three quarters of its view with the next, so the computer can join them.

4 Keep the same height
Fly the whole mission at one height so the map is even.

5 Plan your batteries
Split big areas into smaller missions. Never start a line you can't finish.

6 The instructor holds the controller for automated flights
Students plan, watch and spot. The pilot in command can take over at any time.

7 Check accuracy
Ground markers or RTK GPS make the map line up with the real ground.

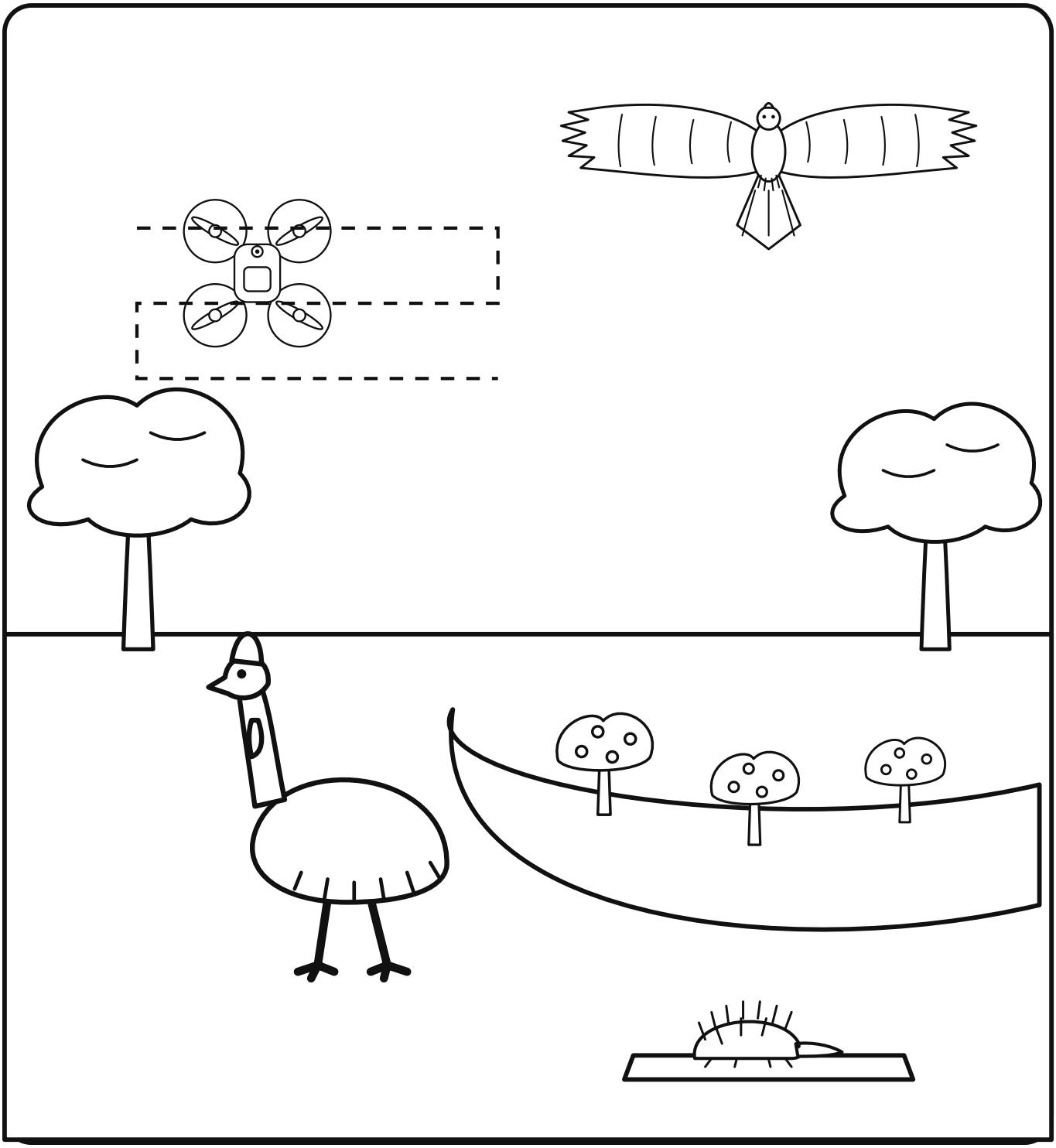
8 Maps belong to Country
Share maps only with the people who own that Country, and keep cultural sites off shared maps.

Rules can change. Check casa.gov.au and a CASA-verified drone safety app before every flight. AI First Nations flights run under ReOC CASA.ReOC.8979.

PAGE 4 · COLOUR & CREATE

GOONDOI'S MAP

Name: _____ Year: _____



Colour the picture with the colours of your Country. **Create:** add one more animal or plant from your Country, and write its name in your language if you know it.

PAGE 5 · MATCH-UP

MAPPING COUNTRY

Name: _____ Year: _____

Draw a line from each item on the left to its match on the right.

MAPPING WORD	WHAT IT MEANS
Grid •	• Many photos joined into one map
Overlap •	• Photos share part of the same view
Orthomosaic •	• Flying up and back in straight lines
RTK GPS •	• A marked point that checks the map lines up
Battery plan •	• Splitting a big area into smaller flights
Ground marker •	• Very exact location, to a few centimetres

Create: write or draw how today's rules help care for Country.

LEVEL 4 QUIZ

Name: _____ Year: _____

Tick one box for each question.

1. What pattern does a mapping drone fly?

☐ Circles ☐ Any way it likes ☐ A grid, up and back in lines

2. Why do photos need to overlap?

☐ So the computer can join them into one map ☐ To use more memory

☐ It looks nice

3. What is an orthomosaic?

☐ A weather report ☐ Many photos joined into one map

☐ A type of drone

4. Who holds the controller during an automated mapping flight in class?

☐ Nobody ☐ The youngest student ☐ The instructor

5. The area is too big for one battery. What do you do?

☐ Split it into smaller missions ☐ Fly faster ☐ Skip the edges

6. Why keep the same height for the whole mission?

☐ It doesn't matter ☐ So the map is even ☐ To save colours

7. Who should the finished map be shared with?

☐ Everyone online ☐ Nobody ever

☐ The people who own that Country

8. What do rangers use maps of pond apple for?

☐ To find and treat the weeds ☐ To plant more pond apple

☐ For fun only

PAGE 7 · FOR TEACHERS

QUIZ ANSWERS

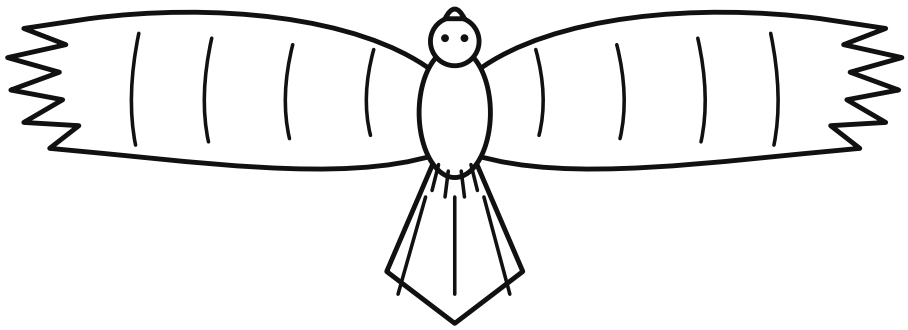
1	A grid, up and back in lines
2	So the computer can join them into one map
3	Many photos joined into one map
4	The instructor
5	Split it into smaller missions
6	So the map is even
7	The people who own that Country
8	To find and treat the weeds

Match-up answers: Grid → Flying up and back in straight lines · Overlap → Photos share part of the same view · Orthomosaic → Many photos joined into one map · RTK GPS → Very exact location, to a few centimetres · Battery plan → Splitting a big area into smaller flights · Ground marker → A marked point that checks the map lines up.

AI FIRST NATIONS · DRONE KNOWLEDGE ON COUNTRY

NGARRAKAN NGARDIJI

LEVEL 4 · MAPPING COUNTRY



This certifies that

_____

has completed Level 4 of Ngarrakan Ngardiji: planning and flying a mapping mission,
following the Civil Aviation Safety Authority (CASA) drone safety rules.

School or community _____ Date _____

_____
John Fejo, Founder & Chief Remote Pilot
AI First Nations · CASA ReOC
CASA.ReOC.8979

_____
Teacher, Elder or ranger